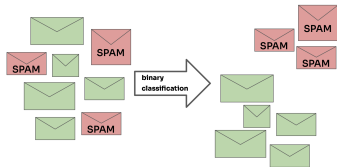
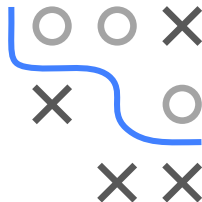


Introduction to Machine Learning

Classification Tasks

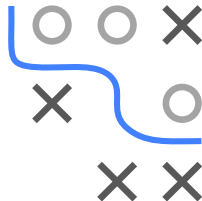


Learning goals

- Classification is supervised learning with categorical labels
- Binary vs. multiclass
- Some examples of classification tasks

CLASSIFICATION

Learn functions that assign class labels to observation / feature vectors.
Each observation belongs to exactly one class. The main difference to regression is that the target is categorical.



Our Data

Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.1	3.5	1.4	0.2	setosa
5.9	3.0	5.1	1.8	virginica



New Data with
unknown label



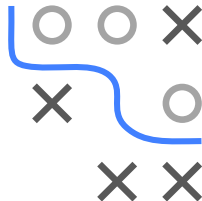
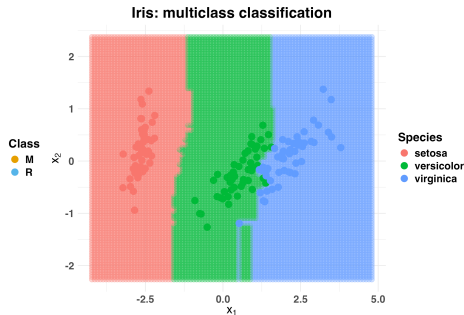
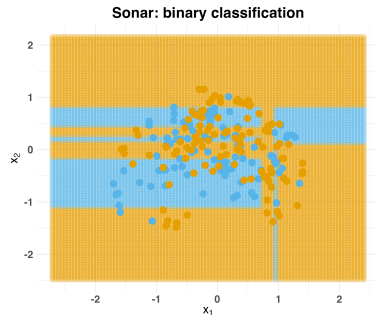
New Class label

Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.4	3.3	3.2	1.1	???

BINARY AND MULTICLASS TASKS

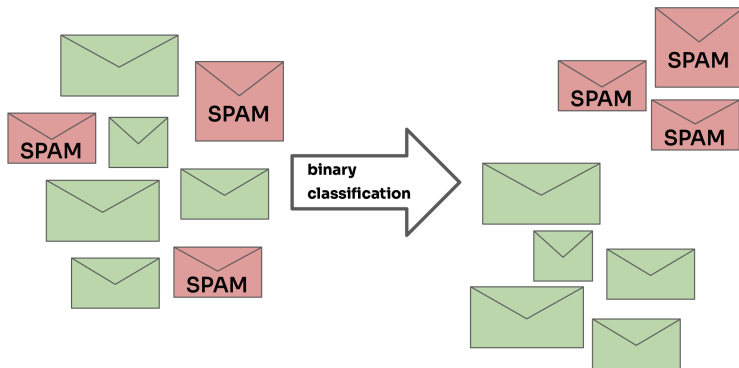
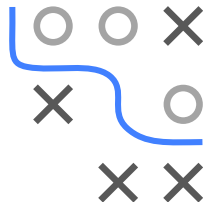
Tasks have a finite number of (unordered) classes.

They can be **binary** or **multiclass**.



BINARY CLASSIFICATION TASK - EXAMPLES

- Credit risk prediction, based on personal data and transactions
- Spam detection, based on textual features
- Churn prediction, based on customer behavior
- Predisposition for specific illness, based on genetic data



MULTICLASS TASK - MEDICAL DIAGNOSIS

INFO

SYMPTOMS

QUESTIONS

CONDITIONS

DETAILS

TREATMENT

Conditions that match your symptoms

UNDERSTANDING YOUR RESULTS ⓘ

Acute Sinusitis

Moderate match

>

Influenza (flu) adults

Moderate match

>

Common cold

Fair match

>

Asthma (Teen and Adult)

Fair match

>

Whooping cough

Fair match

>

↓

LOAD MORE CONDITIONS

Gender **Female** Age **25** Edit

My Symptoms Edit

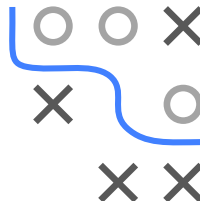
cough, headache, nausea

Could you be pregnant? Edit

No

↺

Start Over



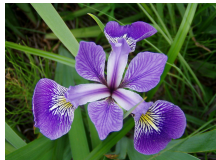
<https://symptoms.webmd.com>

MULTICLASS TASK - IRIS

The iris dataset was introduced by the statistician Ronald Fisher and is one of the most frequent used data sets. Originally, it was designed for linear discriminant analysis.



Setosa



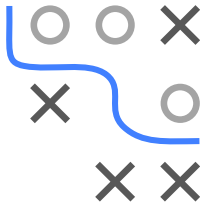
Versicolor



Virginica

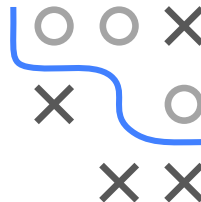
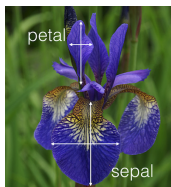
Source:

https://en.wikipedia.org/wiki/Iris_flower_data_set



MULTICLASS TASK - IRIS

- 150 iris flowers
- Predict subspecies
- Based on sepal and petal length / width in [cm]



##		Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
##	1:	5.1	3.5	1.4	0.2	setosa
##	2:	4.9	3.0	1.4	0.2	setosa
##	3:	4.7	3.2	1.3	0.2	setosa
##	4:	4.6	3.1	1.5	0.2	setosa
##	5:	5.0	3.6	1.4	0.2	setosa
##	---					
##	146:	6.7	3.0	5.2	2.3	virginica
##	147:	6.3	2.5	5.0	1.9	virginica
##	148:	6.5	3.0	5.2	2.0	virginica
##	149:	6.2	3.4	5.4	2.3	virginica
##	150:	5.9	3.0	5.1	1.8	virginica

MULTICLASS TASK - IRIS

